Industrial Security (Master)

Solution Sheet – Section 03

Advanced Industrial Protocols

Exercise 3.1 – S7commPlus (mark scheme)

Pass requires: correct version identification (v3 for S7-1500), correct location of the two challenge fields, working KDF in Python that produces the same key as derived in the paper, and a decrypted read-request matching the actual tag name.

Exercise 3.2 – GOOSE Spoof (mark scheme)

The student must show: capture vs. spoof PCAP side-by-side, the subscriber accepting the higher-`stNum` frame, and the only effective mitigation – IEC 62351-6 signed GOOSE (or strict L2 isolation as a workaround).

Exercise 3.3 – Scapy Dissector (mark scheme)

Award marks for: layer that round-trips (encode/decode) at least one captured frame, working replay against a test client, and an honest limits section (no support for fragmentation, no encryption, single endpoint only).

Exercise 3.4 – Fuzz Harness

A clean answer documents: harness, mutation strategy, observed behaviour (crash backtrace, hang, watchdog reset), CWE mapping (CWE-787 OOB write, CWE-400 resource exhaustion, etc.), and a vendor-disclosure note.